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End-to-End Machine Learning Pipeline using Linear Regression

For this task, I used an employee dataset which contains information about employees such as age, experience, education level, location, and salary.

The main objective of this task was to build a complete machine learning pipeline that includes data preprocessing, model training, and evaluation.

First, I loaded the dataset and checked its structure using basic functions like shape, columns, and summary statistics. This helped me understand the data clearly.

Next, I performed basic exploratory data analysis (EDA).

Then , by doing **data preprocessing**. I removed unnecessary columns such as Employee ID and Name as they are not useful for predicting salary. Then, I encoded all categorical variables like education level, location, and gender , so that the data becomes suitable for machine learning.

Then, I split the dataset into **training and testing** sets using an **80-20 ratio**. I applied Linear Regression to train the model using the training data.

After training the model, I **evaluated** its performance using different metrics such as **R^2 score**, **Mean Squared Error (MSE)**, **Mean Absolute Error (MAE)**, and **Root Mean Squared Error (RMSE)**.

I also plotted a graph of actual vs predicted salary to visually check the performance of model.